

Corrigé — Exercice 2

- A. 1)** $f'(x) = \frac{1}{2\sqrt{x-1}} > 0$ sur $]1, +\infty[$: f continue strictement croissante, $f(1) = 2$, $\lim_{+\infty} f = +\infty \Rightarrow$ bijection sur $J = [2, +\infty[$. **2)** $y = \sqrt{x-1} + 2 \Rightarrow (y-2)^2 = x-1$, d'où $f^{-1}(x) = (x-2)^2 + 1$.
- B. 3)** g continue strictement croissante sur $[-1, +\infty[$, $g(-1) = 0$, $\lim_{+\infty} g = +\infty \Rightarrow$ bijection sur $K = [0, +\infty[$. **4)** $g'(x) = \frac{1/2}{2\sqrt{(x+1)/2}} = \frac{1}{4g(x)}$; pour $x \geq 0$, $g(x) \geq g(0) = \frac{1}{\sqrt{2}}$, donc $|g'(x)| \leq \frac{\sqrt{2}}{4}$.
- 5)** $y^2 = \frac{x+1}{2} \Rightarrow g^{-1}(x) = 2x^2 - 1$; $(g^{-1})'(1) = 4$.
- 6)** $f(a) = 4 \iff \sqrt{a-1} = 2 \iff a = 5$; $f'(x) = \frac{1}{2\sqrt{x-1}}$ donne $f'(5) = \frac{1}{4}$, d'où $(f^{-1})'(4) = \frac{1}{f'(5)} = 4$.